

The Four-Coalition International Political Economy

A Mathematical Treatise on

Divine, Capitalist, Labour, and Demoniac Equilibria,
Five-Currency Monetary Hegemony, Military Power, Information, and
Survival-Dominant International Conflict

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This Treatise develops a reduced-order theory of international politics based on four political-economic coalitions—Divine, Capitalist, Labour, and Demoniac—embedded within a monetary, military, institutional, epistemic, technological, and geopolitical system. The construction begins with a dimensional restriction on sovereign complexity: a degree-eight recoverable-money polynomial has nine coefficient dimensions, one of which is allocated to a common recoverable structure while each sovereign unit consumes two informational dimensions. Hence $1 + 2N \leq 9$ and $N_{\max} = 4$. A second restriction comes from the finite-rent theory of reserve currencies. If each viable reserve currency requires at least one-fifth of global reserve rents, then $K_{\max} = 5$.

The resulting international system contains four reduced sovereign political units and five potential reserve-currency nodes. The state-currency architecture is represented by a 4×5 backing matrix. Reserve demand generates seigniorage; seigniorage finances military wages; military wages generate military quality; military quality produces hegemonic capability; and hegemony reinforces reserve demand. Labour enters through a structural labor premium, human-time allocation, bargaining power, capital concentration, human capital, monetary hierarchy, automation, unemployment, inequality, and debt stress. The Divine coalition is represented through legitimacy, sacred authority, identity, and political theology; the Capitalist coalition through capital, finance, property, networks, and institutional accumulation; the Labour coalition through human time, dignity, bargaining power, wages, and participation; and the Demoniac coalition through corruption, information asymmetry, institutional capture, extraction, and epistemic degradation.

The Treatise further introduces a security-regime variable. Ordinary political-economic objectives dominate in peace; military deterrence becomes salient under strategic confrontation; and, when existential conflict crosses a critical threshold, survival becomes the dominant objective. The resulting regime-switching system provides a formal representation of a Divine–Demoniac conflict in which ideological, economic, monetary, and distributive objectives can become subordinate to continued existence. The framework combines economics, finance, political economy, welfare economics, international relations, geopolitics, diplomacy, warfare economics, game theory, statistics, information theory, psychology, philosophy, network science, computer science, artificial intelligence, machine learning, and mathematical modeling.

The paper ends with “The End”

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1 Introduction

International politics is frequently modeled as a collection of sovereign states interacting through trade, diplomacy, military competition, institutions, and ideological conflict. Such a representation is useful but potentially excessive. If the informational structure supporting sovereign decision-making has finite dimensional capacity, then the number of independent sovereign strategic units may itself be bounded.

The starting point of this Treatise is therefore a dimensional reduction.

Let $M(t)$ denote a recoverable-money polynomial of degree d . A degree- d polynomial possesses $d + 1$ coefficients. If one dimension is reserved for the common recoverable structure and every sovereign unit consumes two informational dimensions, feasibility requires

$$1 + 2N \leq d + 1.$$

Hence

$$N \leq \left\lfloor \frac{d}{2} \right\rfloor.$$

For $d = 8$,

$$\boxed{N_{\max} = 4}.$$

The source framework interprets this result as a consequence of dimensional scarcity rather than geography or historical accident.

A second dimensional restriction concerns reserve currencies. Let R_k be reserve demand for currency k and let

$$R^W = \sum_{k=1}^K R_k.$$

Define the reserve share

$$r_k = \frac{R_k}{R^W}.$$

Then

$$\sum_{k=1}^K r_k = 1.$$

If viable reserve status requires

$$r_k \geq \frac{1}{5},$$

then

$$1 = \sum_{k=1}^K r_k \geq \frac{K}{5},$$

so

$$\boxed{K \leq 5}.$$

The two restrictions produce a compact architecture:

$$\boxed{N_{\max} = 4, \quad K_{\max} = 5.}$$

The present Treatise asks what happens when the four sovereign dimensions are interpreted through four political-economic coalitions:

$$\boxed{D = \text{Divine}, \quad C = \text{Capitalist}, \quad L = \text{Labour}, \quad X = \text{Demoniac}.}$$

These four labels are analytical constructs. They are not claims that actual countries or populations are homogeneous members of one ideological category.

2 Source Basis and Model Discipline

The Treatise synthesizes several previously developed frameworks.

Table 1: Principal source frameworks and their roles

Framework	Role in the present Treatise
$N_{\max} \leq 4$	Reduces the independent sovereign state space through finite informational capacity.
Military Wage Seigniorage and Five-Currency Ceiling	Supplies reserve-currency rents, military wage financing, hegemonic feedback, and $K_{\max} \leq 5$.
The Mathematics of Labor	Supplies human-time allocation and the labor-premium framework.
A Theory of Dignified Labor	Supplies $p_l \geq 0$, bargaining power, capital concentration, human capital, monetary hierarchy, automation, unemployment, inequality, and debt stress.
The Manufactured Divine	Supplies the analytical treatment of sacred authority, political theology, legitimacy, identity, and the cost asymmetry of institutionalized divine authority.
Moses: Prophet or Saboteur?	Supplies a model of ideological disruption, labor extraction, adversarial strategy, network diffusion, and reinforcement-learning interpretations of political action.
The Complete Treatise on the Masonic Economy	Supplies the Capitalist coalition's network, transaction-cost, social-capital, property, finance, and institutional mechanisms.
The Complete Treatise on the Demoniac Economy	Supplies corruption, moral hazard, information asymmetry, institutional capture, extraction, and self-reinforcing corruption dynamics.
How Descartes' Demon made Western Philosophy Non-Canonical	Supplies epistemic compression, information asymmetry, canon formation, soft power, and machine-learning interpretations of knowledge selection.
The Standard Nuclear oligARCHy	Supplies nuclear deterrence, coalition stability, crisis detection, redundancy, recapitalization, and survival-oriented security architecture.

The present four-coalition interaction matrix, state-currency affinity system, and survival-dominance regime are theoretical extensions constructed from these primitives. They should therefore be distinguished from empirical estimates.

3 The Dimensional Sovereign Constraint

3.1 Recoverable-Money Polynomial

Let

$$M(t) = \sum_{j=0}^d a_j t^j.$$

The coefficient vector is

$$\mathbf{a} = (a_0, a_1, \dots, a_d)^\top \in \mathbb{R}^{d+1}.$$

A common recoverable structure consumes one informational dimension. Let sovereign unit i consume two dimensions:

$$d_i = 2.$$

Then total required dimensions are

$$D_{\text{required}} = 1 + \sum_{i=1}^N d_i = 1 + 2N.$$

Feasibility requires

$$1 + 2N \leq d + 1.$$

Therefore:

$$N_{\max} = \left\lfloor \frac{d}{2} \right\rfloor.$$

3.2 The Four-Sovereign Bound

For $d = 8$,

$$N_{\max} = \left\lfloor \frac{8}{2} \right\rfloor = 4.$$

Hence:

$$\boxed{N_{\max} = 4.}$$

3.3 Proof

Theorem 1. Suppose a degree- d recoverable-money polynomial has $d + 1$ coefficient dimensions, one common dimension is required, and each sovereign unit requires two informational dimensions. Then

$$N_{\max} = \left\lfloor \frac{d}{2} \right\rfloor.$$

Proof. The degree- d polynomial has $d + 1$ coefficients. Feasibility requires

$$1 + 2N \leq d + 1.$$

Subtracting one gives

$$2N \leq d.$$

Therefore

$$N \leq \frac{d}{2}.$$

Since N is an integer,

$$N_{\max} = \left\lfloor \frac{d}{2} \right\rfloor.$$

□

3.4 Coefficient-Space Diagram

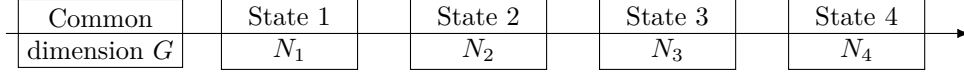


Figure 1: Nine coefficient dimensions for $d = 8$

The decomposition is

$$9 = 1 + 2 + 2 + 2 + 2.$$

The four sovereign dimensions will subsequently be represented by the four coalition states.

4 The Four Political Coalitions

Let

$$\mathbf{q}_i = \begin{pmatrix} d_i \\ c_i \\ l_i \\ x_i \end{pmatrix} \in \Delta^3,$$

where

$$d_i + c_i + l_i + x_i = 1, \quad d_i, c_i, l_i, x_i \geq 0.$$

The global political composition is

$$\mathbf{q} = \begin{pmatrix} d \\ c \\ l \\ x \end{pmatrix} \in \Delta^3.$$

The four components are interpreted as institutional and political modes rather than mutually exclusive demographic identities.

4.1 Divine Coalition

The Divine coalition represents political systems in which legitimacy, sacred authority, identity, transcendence, and moral order play an important role.

Let

$$D = (A_D, I_D, M_D, R_D),$$

where:

- A_D is sacred authority;
- I_D is identity cohesion;
- M_D is moral legitimacy;
- R_D is geopolitical reach.

A reduced objective is

$$U_D = B_D(A_D, I_D, M_D) - C_D(R_D, E_D),$$

where E_D denotes escalation or existential-risk exposure.

The Manufactured Divine framework emphasizes the interaction between sacred authority, political theology, identity, institutional power, and violence. The appropriate analytical implication is therefore not that divine authority is intrinsically beneficial or harmful, but that institutionalized legitimacy can become a source of both coordination and conflict.

4.2 Capitalist Coalition

The Capitalist coalition organizes political economy around capital, property, finance, networks, investment, and institutional accumulation.

Let

$$C = (K, F, N, P),$$

where:

- K is productive and financial capital;
- F is financial depth;
- N is network centrality;
- P is property and institutional ownership.

A reduced objective is

$$U_C = \Pi(K, F) + \lambda_N N + \lambda_P P - \mathcal{R}_C.$$

The Masonic-economy framework is useful here as an institutional-network model: transaction-cost economics, social capital, business networks, voluntary association, property, finance, and governance are all relevant primitives.

4.3 Labour Coalition

The Labour coalition is organized around human time, labor participation, bargaining power, compensation, dignity, employment, and distribution.

Let

$$L = (p_l, f, B, W, H, A, U, I, D),$$

where:

- p_l is the labor premium;
- f is the labor fraction;
- B is bargaining power;
- W is labor compensation;
- H is human capital;
- A is automation intensity;
- U is unemployment;
- I is inequality;
- D is debt stress.

The fundamental dignity constraint is

$$p_l \geq 0.$$

The labor-premium specification is

$$p_l = \alpha B - \beta C + \gamma H.$$

Therefore:

$$\frac{\partial p_l}{\partial B} > 0, \quad \frac{\partial p_l}{\partial C} < 0, \quad \frac{\partial p_l}{\partial H} > 0.$$

The labor fraction is

$$f = \frac{1 + r_f + p_l}{2 + r_f + p_l}$$

and the leisure fraction is

$$g = 1 - f.$$

The condition $p_l \geq 0$ therefore represents simultaneously a dignity condition, a participation condition, a bargaining-power floor, and a social-admissibility condition.

4.4 Demoniac Coalition

The Demoniac coalition is modeled as a theoretical regime in which extraction, corruption, information asymmetry, institutional capture, and moral hazard become systemically reinforcing.

Let

$$X = (Z, H_X, I_X, C_X),$$

where:

- Z is corruption;
- H_X is information asymmetry;
- I_X is institutional capture;
- C_X is coercive or extractive capacity.

A reduced objective is

$$U_X = \lambda_Z Z + \lambda_H H_X + \lambda_I I_X + \lambda_C C_X.$$

The Demoniac Economy supplies a corruption equilibrium in which corruption can generate additional demand through habituation and normalization.

A stylized dynamic is

$$\dot{Z} = aZ + bH_X + cI_X - dR_X,$$

where R_X denotes institutional resistance or reform.

If the reinforcing terms dominate resistance, corruption becomes self-reinforcing.

5 The Epistemic Layer

The international system is not merely material. Information determines which states, ideas, threats, and alternatives are observable.

Let P be the set of all relevant intellectual traditions and $W \subset P$ the institutionally privileged subset. The information contained in the global intellectual set is

$$I_G = \sum_{i \in P} I_i,$$

whereas canonical information is

$$I_C = \sum_{i \in W} I_i.$$

If

$$W \subset P,$$

then generally

$$I_C < I_G.$$

Thus institutional canonization can be interpreted as information compression.

5.1 Question Manifolds

Represent a question by

$$Q \longrightarrow \mathcal{M}_Q,$$

where $\mathcal{M}_Q$ is its manifold of admissible solutions.

A narrow institutional canon reduces

$$\dim(\mathcal{M}_Q).$$

This is analogous to model misspecification and representation bias in machine learning.

If

$$D_W \subset D_G,$$

then training exclusively on D_W can generate:

selection bias, sampling bias, representation bias, epistemic overfitting.

The geopolitical consequence is:

$\text{epistemic power} \rightarrow \text{narrative power} \rightarrow \text{soft power}.$

The financial analogue is portfolio concentration:

$$\Pi = \sum_i w_i I_i.$$

A concentrated epistemic portfolio may maximize institutional consistency while increasing systemic epistemic risk.

6 The Five-Currency Monetary System

Let

$$k \in \{1, 2, 3, 4, 5\}.$$

Let R_k denote reserve demand for currency k and

$$R^W = \sum_{k=1}^5 R_k.$$

Define

$$r_k = \frac{R_k}{R^W}.$$

Then

$$\sum_{k=1}^5 r_k = 1.$$

The viability condition is

$$r_k \geq \frac{1}{5}.$$

6.1 Five-Currency Ceiling

Theorem 2. Suppose:

1. global reserve rents are finite;
2. reserve-currency status requires a minimum level of military wage financing;
3. each viable reserve currency must command at least one-fifth of global reserve rents.

Then

$$K_{\max} = 5.$$

Proof. If every viable currency satisfies

$$r_k \geq \frac{1}{5},$$

then

$$1 = \sum_{k=1}^K r_k \geq \frac{K}{5}.$$

Hence

$$K \leq 5.$$

□

At the exact ceiling,

$$K = 5,$$

all five shares must satisfy

$$r_k = \frac{1}{5}.$$

Thus reserve-share equality at the mathematical ceiling does not imply equal geopolitical power. Differentiation must occur through credibility, institutions, trade, financial depth, military capability, alliances, and technology.

7 The State-Currency Backing Matrix

Define

$$\mathbf{B} = \begin{pmatrix} b_{D1} & b_{D2} & b_{D3} & b_{D4} & b_{D5} \\ b_{C1} & b_{C2} & b_{C3} & b_{C4} & b_{C5} \\ b_{L1} & b_{L2} & b_{L3} & b_{L4} & b_{L5} \\ b_{X1} & b_{X2} & b_{X3} & b_{X4} & b_{X5} \end{pmatrix}.$$

For every state i ,

$$b_{ik} \geq 0, \quad \sum_{k=1}^5 b_{ik} = 1.$$

Thus each row is a currency portfolio.

The architecture is the complete bipartite graph

$$\boxed{K_{4,5}},$$

containing

$$4 \times 5 = 20$$

potential state-currency edges.

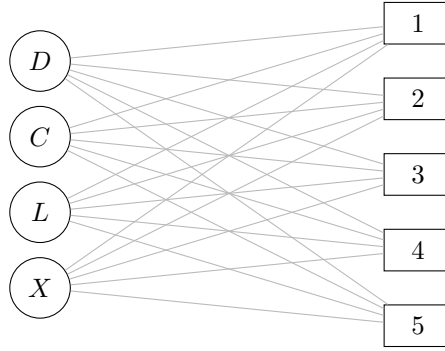


Figure 2: State-currency network

8 Currency Affinity

Define the latent attractiveness of currency k to state i as

$$A_{ik} = \gamma_{Dk}d_i + \gamma_{Ck}c_i + \gamma_{Lk}l_i + \gamma_{Xk}x_i + \mathbf{z}_k^\top \boldsymbol{\beta},$$

where $\mathbf{z}_k$ contains observable currency-issuer characteristics.

The portfolio weight is

$$\boxed{b_{ik} = \frac{\exp(A_{ik})}{\sum_{h=1}^5 \exp(A_{ih})}.$$

This is a multinomial-logit representation of revealed currency affinity.

The affinity matrix is

$$\Gamma = \begin{pmatrix} \gamma_{D1} & \gamma_{D2} & \gamma_{D3} & \gamma_{D4} & \gamma_{D5} \\ \gamma_{C1} & \gamma_{C2} & \gamma_{C3} & \gamma_{C4} & \gamma_{C5} \\ \gamma_{L1} & \gamma_{L2} & \gamma_{L3} & \gamma_{L4} & \gamma_{L5} \\ \gamma_{X1} & \gamma_{X2} & \gamma_{X3} & \gamma_{X4} & \gamma_{X5} \end{pmatrix}.$$

The Γ matrix is an empirical object in a future estimation exercise. The present Treatise specifies its theoretical interpretation but does not claim to have estimated its entries.

9 Military Wage Seigniorage

Let foreign reserve demand be R^W and let s denote the effective seigniorage rate.

For currency k :

$$S_k = sR^W r_k.$$

Suppose a fraction α_k of seigniorage finances military wages.

Then military expenditure is

$$B_k^M = T_k + \alpha_k S_k.$$

If military personnel are M_k , average military compensation is

$$w_k = \frac{T_k + \alpha_k S_k}{M_k}.$$

Military labor supply can be represented by

$$M_k^S = \mathcal{L}\left(\frac{w_k}{w_k^c}\right),$$

with

$$\mathcal{L}' > 0,$$

where w_k^c is civilian compensation.

Military quality satisfies

$$Q_k = Q(M_k, w_k, A_k^{AI}),$$

with

$$\frac{\partial Q_k}{\partial w_k} > 0.$$

Hegemonic capability is

$$H_k = F(Q_k, K_k, N_k),$$

where N_k is alliance-network capability.

Reserve demand is then

$$R_k = G(H_k).$$

Therefore:

$$\boxed{R_k \longrightarrow S_k \longrightarrow w_k \longrightarrow Q_k \longrightarrow H_k \longrightarrow R_k.}$$

9.1 Military Wage-Seigniorage Multiplier

Define

$$\mu_k = \frac{\partial H_k}{\partial S_k}.$$

By the chain rule,

$$\boxed{\mu_k = \frac{\partial H_k}{\partial Q_k} \frac{\partial Q_k}{\partial w_k} \frac{\partial w_k}{\partial S_k}.$$

The total feedback gain is

$$G_k = \frac{\partial R_k}{\partial H_k} \frac{\partial H_k}{\partial Q_k} \frac{\partial Q_k}{\partial w_k} \frac{\partial w_k}{\partial S_k} \frac{\partial S_k}{\partial R_k}.$$

If

$$G_k > 1,$$

the local linearized feedback is self-reinforcing.

Proof. Let δR_k be a small reserve-demand shock. Then:

$$\delta S_k = \frac{\partial S_k}{\partial R_k} \delta R_k.$$

Military wages change by

$$\delta w_k = \frac{\partial w_k}{\partial S_k} \delta S_k.$$

Military quality changes by

$$\delta Q_k = \frac{\partial Q_k}{\partial w_k} \delta w_k.$$

Hegemonic capability changes by

$$\delta H_k = \frac{\partial H_k}{\partial Q_k} \delta Q_k.$$

Finally:

$$\delta R'_k = \frac{\partial R_k}{\partial H_k} \delta H_k.$$

Combining the derivatives gives

$$\delta R'_k = G_k \delta R_k.$$

Therefore $G_k > 1$ implies local amplification. □

10 The Labour–Monetary Connection

The labor premium can depend on institutions, political power, and technology:

$$p_l = p_l(I, P, T).$$

A structural specification is

$$p_l = \alpha B - \beta C + \gamma H.$$

Financial conditions enter through the risk-free rate r_f .

The labor fraction becomes

$$f = \frac{1 + r_f + p_l}{2 + r_f + p_l}.$$

Therefore:

$$\frac{\partial f}{\partial p_l} = \frac{1}{(2 + r_f + p_l)^2} > 0.$$

Thus:

$$p_l \uparrow \Rightarrow f \uparrow.$$

The result is economically important. Human-time allocation is not independent of monetary hierarchy.

10.1 Monetary Hierarchy

Let the sovereign financing condition be

$$r_i^* = r_f - \lambda \Gamma_i - \mu \rho_i,$$

where:

- Γ_i is geopolitical monetary power;
- ρ_i is risk;
- $\lambda, \mu > 0$.

Then:

$$\frac{\partial r_i^*}{\partial \Gamma_i} < 0.$$

Reserve-currency privilege therefore reduces financing costs.
This can influence:

$$r_i^* \rightarrow p_{l,i} \rightarrow f_i \rightarrow U_{L,i}.$$

11 Automation and Labour Stratification

Let A denote automation intensity and S specialized human capital.
A high-skill wage function is

$$W_h = \phi AS.$$

A reduced-form median wage function is

$$W_m = \frac{\omega}{1 + A}.$$

Hence:

$$\frac{\partial W_h}{\partial A} = \phi S > 0,$$

while

$$\frac{\partial W_m}{\partial A} = -\frac{\omega}{(1 + A)^2} < 0.$$

Automation can therefore simultaneously increase returns to specialized labor and reduce median labor compensation.

This produces a bifurcation:

$\text{AI productivity} \not\rightarrow \text{uniform labor empowerment.}$

The corresponding regimes are:

Table 2: Technological and Labour Regimes

Regime	Automation	Bargaining Power	Outcome
Industrial capitalism	Moderate	Strong	Stable labor
Financial capitalism	High	Weakening	Polarization
AI oliGARCHy	Extreme	Low	Stratified society
Post-scarcity	Extreme	Reconstructed	Dignified labor

12 The Four-by-Four Coalition Interaction Matrix

Define

$$\mathbf{A} = \begin{pmatrix} A_{DD} & A_{DC} & A_{DL} & A_{DX} \\ A_{CD} & A_{CC} & A_{CL} & A_{CX} \\ A_{LD} & A_{LC} & A_{LL} & A_{LX} \\ A_{XD} & A_{XC} & A_{XL} & A_{XX} \end{pmatrix}.$$

The ordinary payoff vector is

$$\mathbf{U}^O = \mathbf{A}\mathbf{q}.$$

Thus:

$$U_D^O = A_{DD}d + A_{DC}c + A_{DL}l + A_{DX}x,$$

and similarly for the other coalitions.

A preliminary theoretical sign structure is

	<i>D</i>	<i>C</i>	<i>L</i>	<i>X</i>
<i>D</i>	+	±	±	−
<i>C</i>	±	+	−	±
<i>L</i>	±	−	+	−
<i>X</i>	−	±	−	+

This is a hypothesis-generating matrix, not an empirical finding.

12.1 Divine–Demonic Interaction

The most strongly antagonistic edge is

$$A_{DX} < 0, \quad A_{XD} < 0.$$

The Divine coalition is modeled around legitimacy, moral order, identity, and sacred authority; the Demonic coalition is modeled around extraction, corruption, institutional capture, and information asymmetry.

This interaction can therefore move from ordinary ideological competition to existential conflict.

12.2 Capital–Labour Interaction

The fundamental distributive tension is represented by

$$A_{CL} < 0, \quad A_{LC} < 0,$$

conditional on sufficiently high capital concentration and insufficient bargaining power.

The deeper relationship is

$$C \uparrow \Rightarrow B \downarrow \Rightarrow p_l \downarrow \Rightarrow f \downarrow.$$

This need not hold universally; institutions and human-capital accumulation can reverse the relationship.

12.3 Capital–Demonic Interaction

At moderate levels of corruption, extraction may coexist with capital accumulation. At high levels, corruption can destroy institutional quality and productive capital.

A nonlinear specification is therefore preferable:

$$A_{CX} = a_1Z - a_2Z^2,$$

with an analogous expression for A_{XC} .

The interaction is complementary at low Z and destructive beyond a critical point.

13 Nonlinear Coalition Utility

A more general ordinary utility function is

$$U_i^O = \alpha_i + \mathbf{a}_i^\top \mathbf{q} + \mathbf{q}^\top \mathbf{H}_i \mathbf{q} + \mathbf{z}_i^\top \mathbf{x},$$

where:

- $\mathbf{H}_i$ contains second-order coalition interactions;
- $\mathbf{x}$ contains monetary, technological, military, institutional, labor, and epistemic variables.

This allows the model to represent complementarity, substitution, threshold effects, and nonlinear institutional failure.

14 Welfare Economics

Let social welfare be

$$W^{soc} = \omega_D U_D + \omega_C U_C + \omega_L U_L + \omega_X U_X - \mathcal{E},$$

where $\mathcal{E}$ represents externalities such as:

$$\mathcal{E} = E_{\text{war}} + E_{\text{corruption}} + E_{\text{inequality}} + E_{\text{epistemic}} + E_{\text{environmental}}.$$

A utilitarian planner solves

$$\max_{\mathbf{q}, \mathbf{B}, \mathbf{r}} W^{soc}$$

subject to

$$\mathbf{q} \in \Delta^3,$$

$$\mathbf{r} \in \Delta^4,$$

$$\mathbf{B}\mathbf{1}_5 = \mathbf{1}_4,$$

and

$$p_l \geq 0.$$

The political equilibrium need not maximize social welfare because each coalition internalizes only part of the externality structure.

15 The Institutional Layer

Let institutional quality be

$$I_t \in [0, 1].$$

Regulatory independence in the Demoniac framework satisfies a dynamic of the form

$$\dot{R}(t) = -\lambda I(t)R(t) + \mu(R_0 - R(t)).$$

High corruption pressure reduces institutional independence; reform restores it.
Define institutional resilience as

$$\mathcal{R}_I = \frac{\mu}{\lambda I + \mu}.$$

Then:

$$\frac{\partial \mathcal{R}_I}{\partial I} < 0$$

when I represents corruption pressure.
The broader institutional state affects:

$$\Gamma = \Gamma(\mathcal{I}),$$

so epistemic and institutional quality influence monetary and geopolitical power.

16 Psychology and Behavioral Economics

The Demoniac framework supplies hyperbolic discounting:

$$D(t) = \frac{1}{1 + \alpha t^\beta}.$$

Immediate rewards can therefore be overweighted relative to long-run costs.
Loss aversion generates persistence after initial compromise.
These mechanisms can be incorporated into an individual corruption probability:

$$P(c_i = 1) = \sigma \left(\theta_0 + \theta_1 R_i^{short} - \theta_2 C_i^{future} + \theta_3 Z_i \right).$$

The system-level corruption state then becomes

$$Z_{t+1} = Z_t + \alpha_c A Z_t - \beta_c Z_t + \gamma_c \varepsilon_t.$$

This provides a bridge between microbehavior and institutional dynamics.

17 Network Theory

Let A^{all} be an alliance adjacency matrix.

A centrality vector $\mathbf{c}$ satisfies

$$\mathbf{c} = A^{all} \mathbf{c}.$$

More generally, for a normalized centrality operator:

$$\mathbf{c} = \lambda A^{all} \mathbf{c},$$

where λ is chosen such that the dominant eigenvalue condition holds.
A currency issuer's geopolitical credibility may then be represented as

$$C_k = C(M_k, A_k^{all}, I_k, c_k).$$

Reserve demand becomes

$$R_k = R(C_k, F_k, T_k).$$

Hence:

military power $\rightarrow$ alliance centrality $\rightarrow$ currency credibility $\rightarrow$ reserve demand.

18 International Relations and Diplomacy

The state-currency system produces three distinct forms of credibility:

$$C_i = C_i^{mil} + C_i^{dip} + C_i^{inst}.$$

Military credibility depends on:

$$C_i^{mil} = C(M_i, Q_i, R_i^{res}).$$

Diplomatic credibility depends on:

$$C_i^{dip} = C(A_i^{all}, T_i^{trade}, D_i^{treaty}).$$

Institutional credibility depends on:

$$C_i^{inst} = C(I_i, E_i^{epistemic}, F_i).$$

Reserve demand therefore becomes

$$R_i = G(C_i^{mil}, C_i^{dip}, C_i^{inst}).$$

This formalizes the proposition that reserve currencies are simultaneously economic and geopolitical institutions.

19 Military Economics

Military labor is specialized labor.

Examples include:

- pilots;
- cyber operators;
- naval officers;
- intelligence analysts;
- nuclear technicians;
- autonomous-system operators;
- AI and machine-learning specialists.

Let military compensation be

$$w^M = w^* + \delta,$$

where the reserve-currency privilege produces a wage premium

$$\delta = \phi S.$$

Then:

$$\frac{\partial M^S}{\partial S} > 0.$$

Military wage seigniorage therefore affects recruitment, retention, human capital, and strategic capability.

20 The Security Regime

The political system cannot be represented by a single static game.

Define the existential-threat index

$$\Omega_i = \eta_1 U_i^{unemp} + \eta_2 I_i^{ineq} + \eta_3 D_i^{debt} + \eta_4 M_i^{conf} + \eta_5 E_i^{esc} + \eta_6 N_i^{nuc}.$$

Let the survival-salience function be

$$s_i = \frac{1}{1 + \exp[-\kappa(\Omega_i - \Omega_c)]}.$$

Then:

$$0 \leq s_i \leq 1.$$

Three regimes arise naturally.

Table 3: Security Regimes

Regime	Threat	s_i	Dominant objective
Peace	Low	≈ 0	Political-economic welfare
Deterrence	Intermediate	$0 < s_i < 1$	Mixed political and security objectives
Survival	Extreme	≈ 1	Continued existence

21 Survival-Dominant Utility

Let ordinary utility be U_i^O and survival utility be U_i^S .

Define total utility:

$$U_i = (1 - s_i)U_i^O + s_i U_i^S.$$

Survival utility is

$$U_i^S = P_i^{surv} V_i^{exist} - C_i^M - C_i^E,$$

where:

- P_i^{surv} is survival probability;
- V_i^{exist} is the value of continued existence;
- C_i^M is military mobilization cost;
- C_i^E is escalation cost.

Let

$$P_i^{surv} = \sigma [\kappa_M (M_i - M_i^*) - \kappa_E E_i + \kappa_R R_i^{res}].$$

Military capability and resilience increase survival probability; escalation reduces it.

21.1 Survival-Dominance Proposition

Proposition 3. Suppose

$$U_i^S > U_i^O$$

for sufficiently high existential threat and

$$\lim_{\Omega_i \rightarrow \infty} s_i = 1.$$

Then

$$\lim_{\Omega_i \rightarrow \infty} U_i(\Omega_i) = U_i^S.$$

Proof.
From

$$U_i = (1 - s_i)U_i^O + s_iU_i^S,$$

and

$$\lim_{\Omega_i \rightarrow \infty} s_i = 1,$$

we have

$$\lim_{\Omega_i \rightarrow \infty} (1 - s_i) = 0.$$

Therefore:

$$\lim_{\Omega_i \rightarrow \infty} U_i = 0 \cdot U_i^O + 1 \cdot U_i^S = U_i^S.$$

□

The proposition establishes the formal meaning of survival dominance.

22 Nuclear Deterrence

The SNoG framework supplies a nuclear game in which mutual annihilation has terminal payoff

$$P_{ij} = P_{ji} = -\infty.$$

A stylized nuclear game is:

	C	D
C	(R_{ij}, R_{ji})	(S_{ij}, T_{ji})
D	(T_{ij}, S_{ji})	(P_{ij}, P_{ji})

with

$$T > R > P > S.$$

The terminal state is therefore:

mutual nuclear annihilation $\Rightarrow U = -\infty$.

This creates a strong deterrence force.

23 Deterrence versus Active Conflict

We distinguish:

$$\mathcal{M}_0 = \text{deterrence},$$

$$\mathcal{M}_1 = \text{active conventional or limited conflict},$$

and

$$\mathcal{M}_2 = \text{existential conflict}.$$

The transition is:

$$\mathcal{M}_0 \rightarrow \mathcal{M}_1 \rightarrow \mathcal{M}_2.$$

De-escalation permits:

$$\mathcal{M}_2 \rightarrow \mathcal{M}_1 \rightarrow \mathcal{M}_0.$$

The state is therefore a hybrid dynamical system rather than a single equilibrium.

24 The Divine–Demoniac Conflict

Consider a conflict between Divine and Demoniac systems.

The antagonistic interaction is:

$$A_{DX} < 0, \quad A_{XD} < 0.$$

Define bilateral existential threat:

$$E_{DX} = \chi_D d + \chi_X x + \chi_{DX} M_{DX}.$$

At low threat:

$$E_{DX} < E_1,$$

so:

$$s_D, s_X \approx 0.$$

At deterrence:

$$E_1 \leq E_{DX} < E_2,$$

so:

$$0 < s_D, s_X < 1.$$

At existential conflict:

$$E_{DX} \geq E_2,$$

and:

$$s_D, s_X \rightarrow 1.$$

Consequently:

$$\boxed{U_D \rightarrow U_D^S, \quad U_X \rightarrow U_X^S.}$$

The Divine coalition ceases to optimize only legitimacy and identity, while the Demoniac coalition ceases to optimize only extraction and institutional capture. Both become survival-maximizing systems.

25 Threat-Induced Coalition Compression

Suppose the four coalition utilities are:

$$U_D, \quad U_C, \quad U_L, \quad U_X.$$

Under ordinary conditions, conflict between them can generate fragmentation.

Under existential threat, however,

$$U_i^S(\text{cooperation}) > U_i^S(\text{fragmentation})$$

may hold.

Then previously antagonistic coalitions may form a survival coalition.

For example:

$$D + C + L \longrightarrow \text{survival coalition against } X.$$

The reverse is also mathematically possible.

Thus:

$$\boxed{\text{existential threat} \rightarrow \text{coalition compression.}}$$

This is a general result of the survival-dominance formulation.

26 Capital and Labour Mobilization

During existential conflict, capital becomes partially military capital:

$$K \rightarrow K^M.$$

Labor becomes partially military labor:

$$L \rightarrow L^M.$$

Financial resources become defense resources:

$$R \rightarrow R^M.$$

Industrial output becomes military output:

$$Y \rightarrow Y^M.$$

Hence:

$$\boxed{\text{Survival regime} = \text{capital mobilization} + \text{labor mobilization} + \text{financial mobilization} + \text{military mobilization}.$$

However, labor mobilization remains subject to:

$$p_l \geq 0.$$

Therefore the dignity constraint acts as a sustainability constraint on wartime mobilization.

27 Resilience

Military survival is not equivalent to military expenditure.

Define resilience:

$$R_i^{res} = F(R_i^{red}, C_i^{com}, I_i^{info}, K_i^{cap}, M_i^{mil}),$$

where:

- R_i^{red} is redundancy;
- C_i^{com} is secure communication;
- I_i^{info} is information integrity;
- K_i^{cap} is recapitalization capacity;
- M_i^{mil} is military capability.

The SNoG framework emphasizes distributed decision-making, redundancy, crisis detection, recapitalization, secure communications, and adaptive response.

A stylized survival probability is

$$P_i^{surv} = P_i^{surv}(R_i^{res}, M_i, E_i).$$

This means that survivability is a systems-engineering property rather than a simple function of weapons expenditure.

28 Coalition Stability

Let $\mathcal{C}$ be a possible coalition subset of

$$\{D, C, L, X\}.$$

A generalized coalition-stability condition is:

$$\sum_{i=1}^4 U_i(S) \geq \max_{\mathcal{C}} \left[\sum_{i \in \mathcal{C}} U_i(\mathcal{C}) + \sum_{j \notin \mathcal{C}} U_j(S \setminus \mathcal{C}) \right].$$

This is a cooperative-stability condition.

If the inequality fails, some coalition has an incentive to defect.

Under existential conflict, the utilities must be evaluated using:

$$U_i = (1 - s_i)U_i^O + s_i U_i^S.$$

Thus coalition stability itself is regime dependent.

29 The Complete Multilayer Network

The political layer is:

$$K_4.$$

The monetary layer is:

$$K_{4,5}.$$

The military layer is an alliance and deterrence network.

The epistemic layer is a directed information network.

The full system is therefore a multilayer graph:

$$\mathcal{G} = (V, E_P, E_M, E_A, E_I).$$

where:

- E_P contains political interactions;
- E_M contains state-currency links;
- E_A contains alliance and military links;
- E_I contains information and epistemic links.

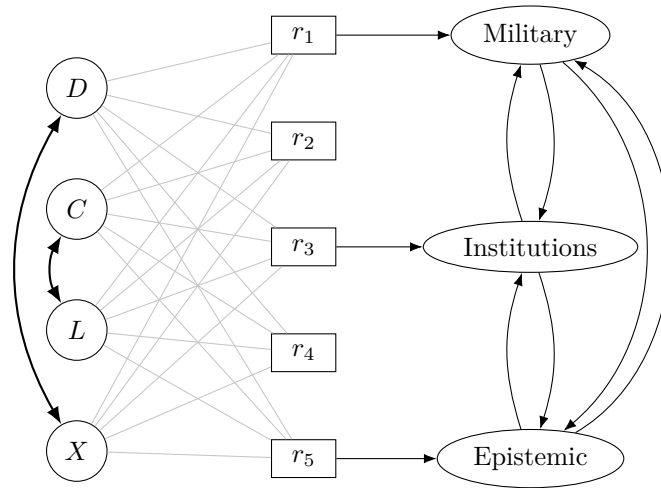


Figure 3: Political-monetary-military-epistemic system

30 The Full State Vector

Define the international state:

$$\mathcal{X}_t = (\mathbf{q}_t, \mathbf{B}_t, \mathbf{r}_t, \mathbf{p}_{l,t}, \mathbf{K}_t, \mathbf{H}_t, \mathbf{M}_t, \mathbf{E}_t, \mathbf{s}_t, \mathcal{I}_t).$$

Its law of motion is

$$\mathcal{X}_{t+1} = F(\mathcal{X}_t, \varepsilon_{t+1}).$$

The system has multiple attractors:

$$\mathcal{A} = \{\mathcal{E}_O, \mathcal{E}_D, \mathcal{E}_S\},$$

where:

$\mathcal{E}_O$ = ordinary political-economic equilibrium,

$\mathcal{E}_D$ = deterrence equilibrium,

and

$\mathcal{E}_S$ = survival equilibrium.

31 Integrated Equilibrium System

The complete model can be summarized by:

$$\begin{aligned} \mathbf{q}^* &= \text{BR}(\mathbf{q}^*, \mathbf{r}^*, \mathbf{M}^*, \mathcal{I}^*), \\ \mathbf{B}^* &= \text{BR}(\mathbf{q}^*, \mathbf{r}^*, \mathcal{I}^*, \mathbf{H}^*), \\ \mathbf{r}^* &= \mathcal{R}(\mathbf{H}^*, \mathbf{F}^*, \mathbf{T}^*, \mathcal{I}^*), \\ \mathbf{S}^* &= sR^W \mathbf{r}^*, \\ \mathbf{w}^* &= \mathcal{W}(\mathbf{S}^*, \mathbf{M}^*), \\ \mathbf{Q}^* &= \mathcal{Q}(\mathbf{w}^*, \mathbf{M}^*, \mathbf{A}^{AI}), \\ \mathbf{H}^* &= \mathcal{H}(\mathbf{Q}^*, \mathbf{K}^*, \mathbf{A}^{all}), \\ \mathbf{E}^* &= \mathcal{E}(\mathbf{M}^*, \mathbf{q}^*, \mathcal{I}^*), \\ \mathbf{s}^* &= \sigma[\kappa(\mathbf{E}^* - \mathbf{E}_c)]. \end{aligned}$$

The payoff system is:

$$\mathbf{U}^* = (\mathbf{1} - \mathbf{s}^*) \odot \mathbf{U}^O + \mathbf{s}^* \odot \mathbf{U}^S.$$

Here $\odot$ denotes elementwise multiplication.

32 Algorithmic Representation

The model can be implemented as a dynamic simulation.

Algorithm: Four-Coalition International Political Economy

INPUT:

```
degree d = 8
four political coalitions D, C, L, X
five potential reserve currencies
initial political state q
initial currency matrix B
initial reserve shares r
```

institutional state I
 military state M
 labor state L
 time horizon T

STEP 1:
 Compute $N_{\max} = \text{floor}(d / 2)$.
 If $N_{\max} < 4$, reduce the number of sovereign strategic units.

STEP 2:
 Enforce $q_i \geq 0$ and $\sum(q_i) = 1$.

STEP 3:
 Compute currency attractiveness:
 $A_{ik} = \text{Gamma}_k' q_i + Z_k' \text{beta}$.

STEP 4:
 Convert attractiveness into currency backing:
 $b_{ik} = \exp(A_{ik}) / \sum_h \exp(A_{ih})$.

STEP 5:
 Compute reserve shares:
 $r_k = R_k / \sum_h R_h$.

STEP 6:
 Compute seigniorage:
 $S_k = s R^W r_k$.

STEP 7:
 Allocate military financing:
 $B_k^M = T_k + \alpha_k S_k$.

STEP 8:
 Compute military wages and military quality:
 $w_k = B_k^M / M_k$
 $Q_k = Q(M_k, w_k, A_k^{AI})$

STEP 9:
 Compute hegemonic capability:
 $H_k = H(Q_k, K_k, A_k^{\text{all}})$

STEP 10:
 Update reserve demand:
 $R_k = R(H_k, F_k, T_k, I_k)$

STEP 11:
 Compute labor premium:
 $p_l = \alpha B - \beta C + \gamma H$

STEP 12:
 Enforce dignified-labor condition:
 if $p_l < 0$:
 classify state as labor-instability regime

STEP 13:
 Compute unemployment, inequality, debt stress,
 military escalation, and existential threat.

```

STEP 14:
  Compute survival salience:
    s_i = logistic(kappa (Omega_i - Omega_c))

STEP 15:
  Compute coalition payoffs:
    U_i = (1-s_i) U_i^0 + s_i U_i^S

STEP 16:
  Update political composition using a best-response,
  replicator, reinforcement-learning, or evolutionary rule.

STEP 17:
  Check coalition stability and deterrence constraints.

STEP 18:
  Repeat until convergence or T is reached.

OUTPUT:
  equilibrium political composition
  equilibrium currency backing
  reserve shares
  military capability
  labor conditions
  institutional state
  security regime
  survival probability

```

33 Replicator Dynamics

A continuous approximation to coalition evolution is:

$$\dot{q}_i = q_i(U_i - \bar{U}),$$

where

$$\bar{U} = \sum_j q_j U_j.$$

For the four coalitions:

$$\dot{d} = d(U_D - \bar{U}),$$

$$\dot{c} = c(U_C - \bar{U}),$$

$$\dot{l} = l(U_L - \bar{U}),$$

and

$$\dot{x} = x(U_X - \bar{U}).$$

The simplex is invariant because

$$\dot{d} + \dot{c} + \dot{l} + \dot{x} = \sum_i q_i U_i - \bar{U} \sum_i q_i = \bar{U} - \bar{U} = 0.$$

Thus:

$$d + c + l + x = 1$$

is preserved.

34 Reinforcement Learning Interpretation

Each sovereign coalition can alternatively be modeled as an agent in a Markov decision process

$$(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma).$$

The state is

$$\mathcal{S} = \mathcal{X}_t.$$

Actions include:

- monetary allocation;
- alliance formation;
- diplomatic action;
- labor policy;
- capital allocation;
- military mobilization;
- de-escalation;
- information disclosure;
- institutional reform.

The Bellman equation is

$$V_i^*(s) = \max_{a \in \mathcal{A}_i} \left[R_i(s, a) + \gamma \sum_{s'} P(s'|s, a) V_i^*(s') \right].$$

The survival regime changes the reward function:

$$R_i(s, a) = (1 - s_i) R_i^O(s, a) + s_i R_i^S(s, a).$$

Thus an AI agent trained under normal conditions may rationally change its policy when the environment crosses the existential threshold.

35 Causal Identification

The reduced-form relationship

$$R_{it} = \alpha + \beta_1 M_{it} + \beta_2 W_{it} + \beta_3 A_{it} + \beta_4 F_{it} + \varepsilon_{it}$$

is useful descriptively but does not automatically establish causality.

The structural causal graph is:

reserve demand $\rightarrow$ seigniorage $\rightarrow$ military wages $\rightarrow$ military quality $\rightarrow$ hegemony $\rightarrow$ reserve demand.

Identification therefore requires either temporal structure, valid instruments, natural experiments, structural restrictions, or combinations of these.

For an instrument Z for military wages:

$$w_{it} = \pi_0 + \pi_1 Z_{it} + \mathbf{X}_{it}^\top \boldsymbol{\pi} + \nu_{it}.$$

The second stage is

$$H_{it} = \alpha + \beta \hat{w}_{it} + \mathbf{X}_{it}^\top \boldsymbol{\gamma} + \varepsilon_{it}.$$

Relevance requires

$$\text{Cov}(Z, w) \neq 0,$$

while exclusion requires that Z affect H only through the endogenous channel, conditional on controls.

36 Statistical Estimation

The state-currency affinity model can be estimated by multinomial maximum likelihood.

For observation i :

$$\Pr(B_{ik} = 1) = \frac{\exp(A_{ik})}{\sum_h \exp(A_{ih})}.$$

The log-likelihood is

$$\ell(\theta) = \sum_i \sum_k y_{ik} \log b_{ik},$$

where

$$y_{ik} \in \{0, 1\}.$$

For continuous currency portfolios, a compositional-data likelihood can be used.

A Bayesian version is:

$$p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta)p(\theta).$$

Posterior backing is:

$$\hat{b}_{ik} = E[b_{ik}|\mathcal{D}].$$

The dominant currency is

$$k_i^* = \arg \max_k \hat{b}_{ik}.$$

37 Currency Dependence and Diversification

Define currency concentration:

$$D_i = \max_k b_{ik}.$$

Define entropy:

$$H_i^B = - \sum_{k=1}^5 b_{ik} \ln b_{ik}.$$

High D_i means concentrated dependence.

High H_i^B means diversified currency backing.

The two measures are complementary.

38 Epistemic Concentration Risk

Let epistemic weights be w_i .

Define:

$$H_E = - \sum_i w_i \ln w_i.$$

Low H_E indicates concentration.

High H_E indicates diversification.

A portfolio analogy is:

$$\Pi_E = \sum_i w_i I_i.$$

Systemic epistemic risk can be represented as

$$R_E = \text{Var}(\Pi_E) + \lambda(I_G - I_C).$$

The purpose is not to treat all knowledge as numerically identical, but to model the structural consequences of systematic exclusion.

39 Physics Analogy: Phase Transitions

The security regime resembles a statistical-mechanical phase transition.

Let Ω be the order parameter's driver.

Define:

$$s = \sigma[\kappa(\Omega - \Omega_c)].$$

As κ becomes large, the transition approaches a step function:

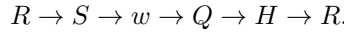
$$s \rightarrow \begin{cases} 0, & \Omega < \Omega_c, \\ 1, & \Omega > \Omega_c. \end{cases}$$

The analogy is useful because a small change in threat can produce a large change in strategic behavior near the critical threshold.

The analogy is not a claim that international politics literally obeys an equilibrium statistical-mechanics Hamiltonian.

40 Chemistry Analogy: Catalysis and Feedback

The reserve-currency mechanism resembles catalytic amplification:



The reserve currency acts as a source of purchasing power that enables the production of military and geopolitical capability, which then increases the attractiveness of the currency.

The analogy becomes unstable if feedback gain exceeds unity:

$$G > 1.$$

A stabilizing policy must therefore reduce at least one component of the product:

$$G = G_R G_H G_Q G_w G_S.$$

This can be interpreted as a policy-design decomposition.

41 Biological Analogy: Adaptation and Selection

Coalition dynamics resemble evolutionary selection.

Let fitness be

$$F_i = U_i.$$

Then replicator dynamics give:

$$\dot{q}_i = q_i(F_i - \bar{F}).$$

Institutions, currencies, technologies, and alliances can therefore be interpreted as phenotypic traits whose prevalence changes with relative strategic fitness.

The analogy is limited: political agents possess intentionality, institutions, language, and reflexivity, unlike biological populations.

42 Philosophical Interpretation

The model contains four distinct philosophical problems.

42.1 Legitimacy

The Divine coalition asks:

Why should authority be obeyed?

42.2 Property

The Capitalist coalition asks:

Who controls productive assets and their returns?

42.3 Labour

The Labour coalition asks:

What compensation makes human-time allocation dignified?

42.4 Epistemic Power

The Demoniac and epistemic layers ask:

Who controls the information upon which choices are made?

The international system therefore contains simultaneous struggles over:

authority, property, labor, information.
--

43 A General Four-Coalition Objective

Let

$$\mathbf{q} = (d, c, l, x).$$

Define the general objective:

$U_i = U_i(\mathbf{q}, \mathbf{r}, \mathbf{B}, p_l, K, W, Z, I, Q, H, M, E).$

The international equilibrium is a fixed point:

$\mathcal{X}^* = F(\mathcal{X}^*).$

The equilibrium may be unique, multiple, unstable, or path dependent.

44 Stability

Let

$$\dot{\mathcal{X}} = F(\mathcal{X}).$$

At equilibrium $\mathcal{X}^*$:

$$F(\mathcal{X}^*) = 0.$$

Linearize:

$$\delta \dot{\mathcal{X}} = J(\mathcal{X}^*) \delta \mathcal{X},$$

where

$$J = \left. \frac{\partial F}{\partial \mathcal{X}} \right|_{\mathcal{X}^*}.$$

The equilibrium is locally asymptotically stable if all eigenvalues satisfy

$$\text{Re}(\lambda_j(J)) < 0.$$

A regime transition can occur when an eigenvalue crosses the imaginary axis.
Thus the mathematical notion of crisis is:

$$\boxed{\max_j \text{Re}(\lambda_j(J)) \rightarrow 0.}$$

This provides a bridge between stability theory, early-warning systems, and international crisis detection.

45 Early-Warning System

Define a systemic-risk index

$$\mathcal{R}_t = \omega_1 U_t + \omega_2 I_t + \omega_3 D_t + \omega_4 E_t + \omega_5 N_t + \omega_6 (1 - I_t^{inst}).$$

A regime transition is predicted when:

$$\mathcal{R}_t > \mathcal{R}_c.$$

A machine-learning classifier may estimate:

$$P(m_{t+1} = m_2 | \mathbf{X}_t) = \sigma(\mathbf{w}^\top \mathbf{X}_t).$$

A recurrent or state-space model may instead estimate:

$$P(m_{t+h} = 2 | \mathbf{X}_{0:t}).$$

The model can therefore be empirically evaluated as an early-warning framework.

46 Monte Carlo Simulation

For parameters θ , simulate:

$$\mathcal{X}_{t+1} = F(\mathcal{X}_t, \varepsilon_{t+1}; \theta).$$

Repeat for

$$b = 1, \dots, B.$$

Estimate:

$$\hat{P}(\mathcal{E}_S) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\mathcal{E}_S^{(b)}\}.$$

Estimate mean survival probability:

$$\hat{P}^{surv} = \frac{1}{B} \sum_{b=1}^B P^{surv, (b)}.$$

Estimate regime-transition probability:

$$\hat{P}_{O \rightarrow S} = \frac{\sum_b \mathbf{1}\{\mathcal{E}_O^{(b)} \rightarrow \mathcal{E}_S^{(b)}\}}{B}.$$

47 Sensitivity Analysis

For a parameter θ_j , define local sensitivity:

$$S_j = \frac{\partial Y}{\partial \theta_j} \frac{\theta_j}{Y}.$$

Global sensitivity may use variance decomposition:

$$S_j^{Sobol} = \frac{\text{Var}_{\theta_j}[E(Y|\theta_j)]}{\text{Var}(Y)}.$$

Important parameters include:

$$\alpha, \beta, \gamma, \lambda, \mu, \kappa, \Omega_c, G, \eta_1, \dots, \eta_6.$$

48 Comparative Statics

The model generates several qualitative predictions.

48.1 Capital Concentration

If

$$\frac{\partial C}{\partial K} > 0$$

and

$$\frac{\partial p_l}{\partial C} < 0,$$

then:

$$\frac{\partial p_l}{\partial K} < 0$$

unless bargaining power or human capital offsets the effect.

48.2 Bargaining Power

Since

$$\frac{\partial p_l}{\partial B} > 0,$$

stronger labor bargaining increases labor participation:

$$\frac{\partial f}{\partial B} > 0.$$

48.3 Reserve-Currency Privilege

Since

$$S_k = sR^W r_k,$$

we obtain:

$$\frac{\partial S_k}{\partial r_k} = sR^W > 0.$$

If

$$\frac{\partial H_k}{\partial S_k} > 0,$$

then:

$$\frac{\partial H_k}{\partial r_k} > 0.$$

48.4 Military Feedback

If

$$G_k > 1,$$

then a positive reserve-demand shock is locally amplified.

If

$$G_k < 1,$$

the feedback is locally damped.

49 Policy Instruments

The system has several policy levers.

Table 4: Policy Instruments by Layer

Layer	Instrument	Primary effect
Political	Institutional reform	Reduce coalition externalities
Labour	Bargaining institutions	Increase B and p_l
Finance	Macroprudential policy	Reduce systemic financial risk
Monetary	Reserve diversification	Reduce currency concentration
Military	Deterrence	Reduce existential conflict probability
Military	Redundancy	Increase survival probability
Information	Epistemic diversification	Reduce information concentration
Technology	Human-capital investment	Increase H and adaptive capacity
Diplomacy	Alliance management	Increase credibility and reduce escalation
Governance	Anti-corruption	Reduce Z and institutional capture
AI	Model diversification	Reduce representation and selection bias

50 A Minimum-Survival Problem

Suppose the state chooses policy vector

$$\mathbf{a} = (a^{mil}, a^{lab}, a^{fin}, a^{dip}, a^{inst}).$$

The state solves:

$$\min_{\mathbf{a}} C(\mathbf{a})$$

subject to

$$P^{surv}(\mathbf{a}) \geq \bar{P},$$

$$p_l(\mathbf{a}) \geq 0,$$

and

$$I^{inst}(\mathbf{a}) \geq \bar{I}.$$

The Lagrangian is

$$\mathcal{L} = C(\mathbf{a}) + \lambda_P(\bar{P} - P^{surv}) + \lambda_l(-p_l) + \lambda_I(\bar{I} - I^{inst}).$$

The Kuhn–Tucker conditions characterize the minimum-cost survival architecture.

51 Existential Conflict as an Objective-Function Transformation

The central conceptual result of the Treatise can now be stated compactly.

In peace:

$$U_i \approx U_i^O.$$

Under deterrence:

$$U_i = (1 - s_i)U_i^O + s_iU_i^S, \quad 0 < s_i < 1.$$

Under existential conflict:

$$U_i \approx U_i^S.$$

Thus:

security conditions transform the objective function itself.

This is stronger than simply adding military expenditure to a conventional political economy model.

52 The Complete Causal Architecture

The full system can be represented as:

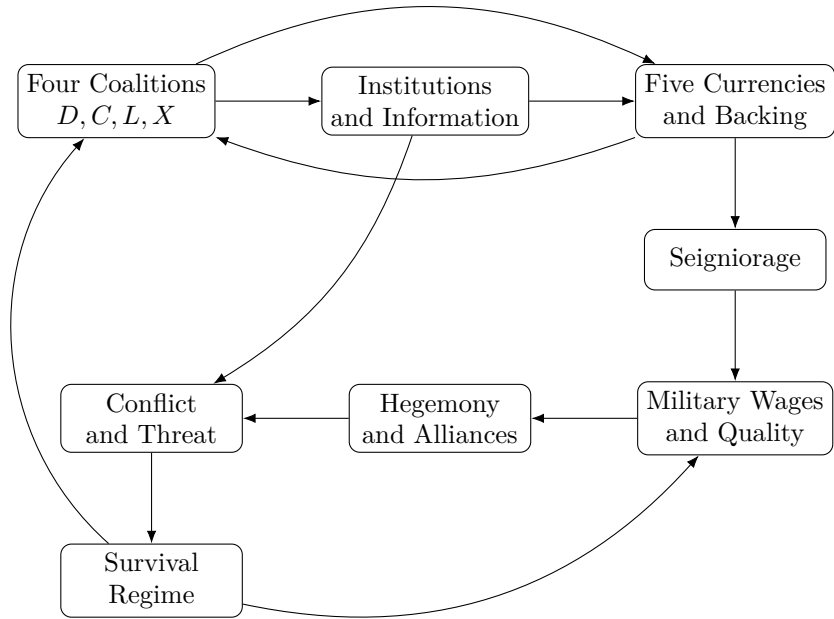


Figure 4: The complete causal architecture

The core loop is:

Politics → Institutions → Currency → Seigniorage → Military Power → Hegemony → Reserve Demand → Politics.

The crisis loop is:

Inequality + Debt + Unemployment + Institutional Failure + Military Escalation → Survival Salience.

53 Master Equations

The Treatise can finally be summarized by the following system:

$$N_{\max} = \left\lfloor \frac{d}{2} \right\rfloor = 4.$$

$$K_{\max} = 5.$$

$$\mathbf{q} \in \Delta^3.$$

$$\mathbf{r} \in \Delta^4.$$

$$\mathbf{B} \in \mathbb{R}_+^{4 \times 5}, \quad \mathbf{B}\mathbf{1}_5 = \mathbf{1}_4.$$

$$S_k = sR^W r_k.$$

$$R_k \rightarrow S_k \rightarrow w_k \rightarrow Q_k \rightarrow H_k \rightarrow R_k.$$

$$p_l = \alpha B - \beta C + \gamma H, \quad p_l \geq 0.$$

$$f = \frac{1 + r_f + p_l}{2 + r_f + p_l}.$$

$$\mathbf{U}^O = \mathbf{A}\mathbf{q}$$

and

$$U_i = (1 - s_i)U_i^O + s_iU_i^S.$$

Finally,

$$s_i = \frac{1}{1 + \exp[-\kappa(\Omega_i - \Omega_c)]}.$$

Therefore:

$$\mathcal{X}_{t+1} = F(\mathcal{X}_t, \varepsilon_{t+1}).$$

54 Principal Theoretical Results

The principal results are:

1. Finite informational capacity implies a finite sovereign state space.
2. For a degree-eight recoverable-money polynomial, the bound is $N_{\max} = 4$.
3. Finite reserve rents and a 20-percent viability threshold imply $K_{\max} = 5$.
4. The resulting state-currency architecture contains at most twenty potential state-currency links.
5. Reserve-currency rents can finance specialized military labor.
6. Military wages can increase military quality and therefore geopolitical capability.
7. Geopolitical capability can reinforce reserve-currency demand.
8. The feedback loop is locally self-reinforcing when $G > 1$.

9. Labour has a structural dignity constraint $p_l \geq 0$.
10. Capital concentration reduces the labor premium unless offset by bargaining power or human capital.
11. Automation can simultaneously increase returns to specialized labor and reduce median labor compensation.
12. Corruption and institutional capture can become self-reinforcing.
13. Epistemic concentration can reduce the effective solution space available to a political system.
14. Political, monetary, military, institutional, and epistemic networks are jointly endogenous.
15. Security conditions can transform the objective function itself.
16. Under sufficiently high existential threat, survival becomes asymptotically dominant.
17. Formerly antagonistic coalitions can form temporary survival coalitions when the cost of fragmentation exceeds the cost of cooperation.

55 Limitations and Identification Boundaries

The model is deliberately multidisciplinary, but several distinctions must be preserved.

First, the four labels D, C, L, X are analytical ideal types. The source papers do not establish an empirical classification of contemporary states into these four categories.

Second, the $N_{\max} \leq 4$ result is conditional upon the dimensional assumptions of the recoverable-money framework.

Third, the Five-Currency Ceiling is conditional upon the finite-rent and one-fifth minimum-viability assumptions.

Fourth, the military wage-seigniorage loop is a structural hypothesis. A regression showing correlation between military wages and reserve status does not by itself prove the entire causal chain.

Fifth, the Divine and Demoniatic categories are theoretical constructs and should not be interpreted as empirical judgments about religious believers or real populations.

Sixth, the physics, chemistry, biology, and evolutionary analogies are mathematical analogies rather than claims of literal physical identity.

Seventh, the survival regime is an extension of the deterrence framework. The assumption that survival dominates ordinary utility under existential threat is a deliberate modeling choice.

These limitations do not invalidate the framework. They define the empirical work required to test it.

56 Research Program

The next empirical stage should estimate:

$$\Gamma, \quad \mathbf{B}, \quad \mathbf{r}, \quad \mathbf{A}, \quad G, \quad \Omega_c.$$

A global panel can then be constructed with:

- reserve-currency shares;
- central-bank reserve composition;
- military expenditure;
- military wage bills;
- military personnel;
- military human capital;
- alliance centrality;

- trade invoicing;
- financial depth;
- capital concentration;
- labor bargaining power;
- labor participation;
- unemployment;
- inequality;
- debt stress;
- automation;
- institutional quality;
- corruption;
- information and media concentration;
- geopolitical conflict;
- crisis and escalation indicators.

The estimation architecture should combine:

$$\boxed{\text{panel econometrics} + \text{causal inference} + \text{network analysis} + \text{state-space models} + \text{Bayesian estimation} + \text{machine learning} + \text{agent-based simulation} .}$$

57 Conclusion

The Four-Coalition International Political Economy can be summarized as a reduced multilayer system:

$$\boxed{\underbrace{N_{\max} \leq 4}_{\text{sovereign dimensional capacity}} + \underbrace{K_{\max} \leq 5}_{\text{monetary capacity}} + \underbrace{D, C, L, X}_{\text{political-economic modes}} + \underbrace{\text{Peace, Deterrence, Survival}}_{\text{security regimes}} .}$$

The four coalitions govern ordinary political-economic competition.

The five currencies provide the monetary architecture.

Seigniorage connects reserve status to military labor.

Military labor generates military quality.

Military quality generates geopolitical capability.

Geopolitical capability reinforces reserve demand.

Labour theory places human time and dignity inside the system.

Institutional and epistemic theory determines how information and authority are processed.

Corruption theory describes a competing self-reinforcing regime of extraction and institutional degradation.

Nuclear deterrence imposes a catastrophic terminal payoff on uncontrolled escalation.

And the survival function provides the final regime transformation:

$$\boxed{\text{ordinary utility} \longrightarrow \text{deterrence utility} \longrightarrow \text{survival utility} .}$$

The central insight is therefore not that Divine, Capitalist, Labour, or Demoniatic politics permanently dominates the others.

Rather:

$$\boxed{\text{international politics is a regime-switching contest over authority, capital, labor, information, money, military power, and ultimately survival.}}$$

When existential threat is low, the four coalitions compete over ordinary political and economic objectives.

When existential threat becomes substantial, military and monetary constraints become increasingly important.

When existential threat becomes sufficiently large,

$$s_i \rightarrow 1,$$

and the objective function itself changes:

$$U_i \rightarrow U_i^S.$$

Thus a conflict between Divine and Demoniatic systems can begin as a contest over legitimacy, identity, institutions, information, and political order, pass through deterrence and military mobilization, and ultimately enter a regime in which the dominant strategic variable is no longer wealth, ideology, labor, capital, or institutional power, but the probability of continued existence.

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Glossary

D	Divine coalition; an analytical political-economic regime emphasizing sacred authority, legitimacy, identity, moral order, and transcendence.
C	Capitalist coalition; an analytical regime emphasizing capital, property, finance, investment, networks, and institutional accumulation.
L	Labour coalition; an analytical regime emphasizing human time, labor, bargaining power, dignity, employment, compensation, and distribution.
X	Demoniac coalition; an analytical regime emphasizing corruption, extraction, information asymmetry, institutional capture, and moral degradation.
$N_{\max}$	Maximum number of independent sovereign strategic units permitted by the dimensional architecture.
$K_{\max}$	Maximum number of simultaneously viable reserve currencies under the finite-rent viability assumptions.
p_l	Labor premium; minimum structural compensation required for dignified labor participation beyond leisure.
f	Labor fraction; fraction of available human time devoted to labor.
g	Leisure fraction; fraction of available human time devoted to leisure.
B	Labor bargaining power.
C	Capital concentration when used as a variable inside the labor-premium equation.
H	Human capital or hegemonic capability depending on context; subscripts should be used to avoid ambiguity.

A	Automation intensity.
R_k	Reserve demand for currency k .
R^W	World reserve demand.
r_k	Global reserve share of currency k .
S_k	Seigniorage associated with currency k .
w_k	Military wage associated with currency issuer k .
Q_k	Military quality.
H_k	Hegemonic capability.
G_k	Military wage-seigniorage feedback gain.
$\mathbf{B}$	4×5 state-currency backing matrix.
b_{ik}	Currency backing of state i by currency k .
Γ	Four-by-five coalition-currency affinity matrix.
$\mathbf{q}$	Political composition vector in the four-dimensional coalition simplex.
$\mathbf{r}$	Reserve-currency share vector.
s_i	Survival salience of sovereign unit i .
Ω_i	Existential-threat index.
Ω_c	Critical threat threshold.
U_i^O	Ordinary political-economic utility.
U_i^S	Survival utility.
U_i	Regime-dependent total utility.
P_i^{surv}	Probability of survival.
V_i^{exist}	Value assigned to continued existence.
R_i^{res}	Resilience capacity.
Z	Corruption.
I	Institutional quality, or inequality when used with an appropriate superscript.
H_X	Information asymmetry in the Demoniac subsystem.
I_X	Institutional capture in the Demoniac subsystem.
$K_{4,5}$	Complete bipartite graph connecting four sovereign nodes to five currency nodes.
$\mathcal{X}_t$	Full international state vector at time t .
$\mathcal{E}_O$	Ordinary political-economic equilibrium.
$\mathcal{E}_D$	Deterrence equilibrium.
$\mathcal{E}_S$	Survival equilibrium.
Survival dominance	
The condition in which survival utility becomes the dominant component of total utility.	

Deterrence

A strategic regime in which catastrophic retaliation makes uncontrolled escalation unattractive.

Epistemic compression

Reduction in the accessible information or solution space caused by institutional selection.

Question manifold

The set of admissible solutions generated by a formulation of a question.

Currency backing

The extent to which a political state relies on a reserve currency for monetary, financial, strategic, or geopolitical support.

Military wage-seigniorage loop

The feedback $R \rightarrow S \rightarrow w \rightarrow Q \rightarrow H \rightarrow R$.

Coalition compression

Reduction in political fragmentation caused by a sufficiently large common existential threat.

Dignified labor

Labor supplied under the condition $p_l \geq 0$, so that human time retains irreducible structural value.

AI oliGARCHy

A technological regime in which computational infrastructure, productive capacity, or high-skill human capital become highly concentrated.

Warfare economics

Analysis of the interaction between economic resources, military capability, incentives, and geopolitical conflict.

Resilience

The capacity of a political-economic system to absorb shocks while maintaining essential functions.

The End